In [1]:
```
import pandas as pd
```

In [2]:
```
import numpy as np
```

In [3]:
```
import random as rnd
```

In [4]:
```
import seaborn as sns
```

In [5]:
```
import matplotlib.pyplot as plt
```

In [6]:
```
%matplotlib inline
```

In [7]:
```
data = pd.read_csv("/content/train_and_test2.csv")
```

In [8]:
```
data
```

Out[8]:

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	zer
0	1	22.0	7.2500	0	1	0	0	0	0	
1	2	38.0	71.2833	1	1	0	0	0	0	
2	3	26.0	7.9250	1	0	0	0	0	0	
3	4	35.0	53.1000	1	1	0	0	0	0	
4	5	35.0	8.0500	0	0	0	0	0	0	
...	...	...	...	...	...	...	...	...	...	
1304	1305	28.0	8.0500	0	0	0	0	0	0	
1305	1306	39.0	108.9000	1	0	0	0	0	0	
1306	1307	38.5	7.2500	0	0	0	0	0	0	
1307	1308	28.0	8.0500	0	0	0	0	0	0	
1308	1309	28.0	22.3583	0	1	0	0	0	0	

1309 rows × 28 columns

In [9]:
```
data.shape
```

Out[9]: (1309, 28)

In [10]:
```
data.describe()
```

Out[10]:

	Passengerid	Age	Fare	Sex	sibsp	zero
count	1309.000000	1309.000000	1309.000000	1309.000000	1309.000000	1309.0
mean	655.000000	29.503186	33.281086	0.355997	0.498854	0.0
std	378.020061	12.905241	51.741500	0.478997	1.041658	0.0
min	1.000000	0.170000	0.000000	0.000000	0.000000	0.0
25%	328.000000	22.000000	7.895800	0.000000	0.000000	0.0
50%	655.000000	28.000000	14.454200	0.000000	0.000000	0.0
75%	982.000000	35.000000	31.275000	1.000000	1.000000	0.0
max	1309.000000	80.000000	512.329200	1.000000	8.000000	0.0

8 rows × 28 columns

In [14]:

```
data.isnull().sum()
```

Out[14]:

	0
Passengerid	0
Age	0
Fare	0
Sex	0
sibsp	0
zero	0
zero.1	0
zero.2	0
zero.3	0
zero.4	0
zero.5	0
zero.6	0
Parch	0
zero.7	0
zero.8	0
zero.9	0
zero.10	0
zero.11	0
zero.12	0
zero.13	0
zero.14	0
Pclass	0
zero.15	0
zero.16	0
Embarked	2
zero.17	0
zero.18	0
2urvived	0

dtype: int64

In [15]:

```
data['Age'] = data['Age'].fillna(np.mean(data['Age']))
```

In [17]:
```
data['Fare'] = data['Fare'].fillna(data['Fare'].mode()[0])
```

In [20]:
```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

In [21]:
```
data.isnull().sum()
```

Out[21]:

	0
Passengerid	0
Age	0
Fare	0
Sex	0
sibsp	0
zero	0
zero.1	0
zero.2	0
zero.3	0
zero.4	0
zero.5	0
zero.6	0
Parch	0
zero.7	0
zero.8	0
zero.9	0
zero.10	0
zero.11	0
zero.12	0
zero.13	0
zero.14	0
Pclass	0
zero.15	0
zero.16	0
Embarked	0
zero.17	0
zero.18	0
2urvived	0

dtype: int64

In [22]:

```
data[['Sex', 'Age']].groupby(['Sex'], as_index=False).mean().sort_values(by='A
```

Out[22]:

	Sex	Age
0	0	30.017888
1	1	28.572082

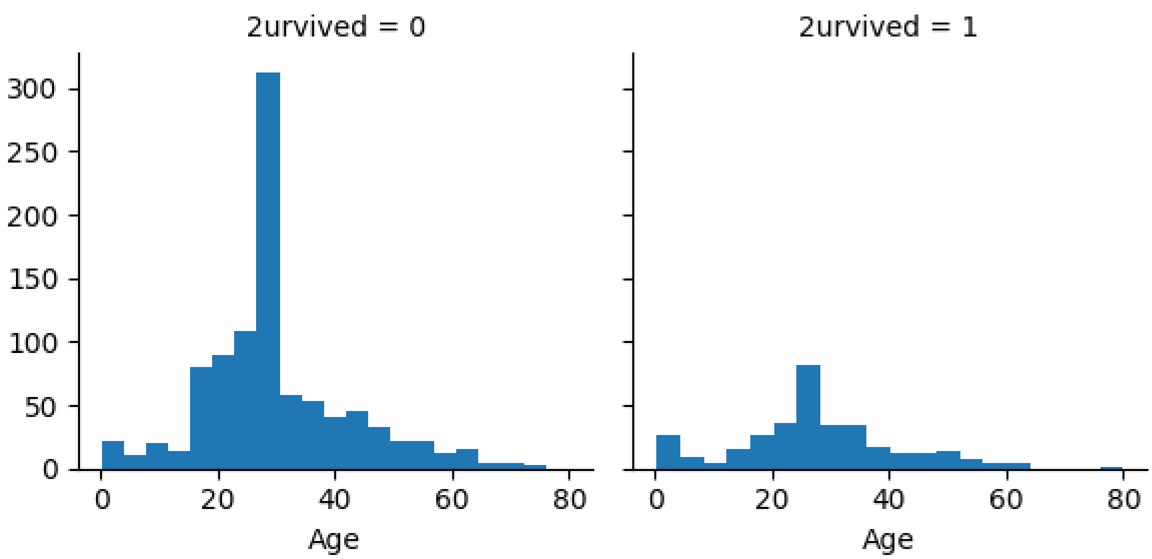
In [25]:
```
data[["sibsp", "2urvived"]].groupby(['sibsp'], as_index=False).mean().sort_val
```

Out[25]:

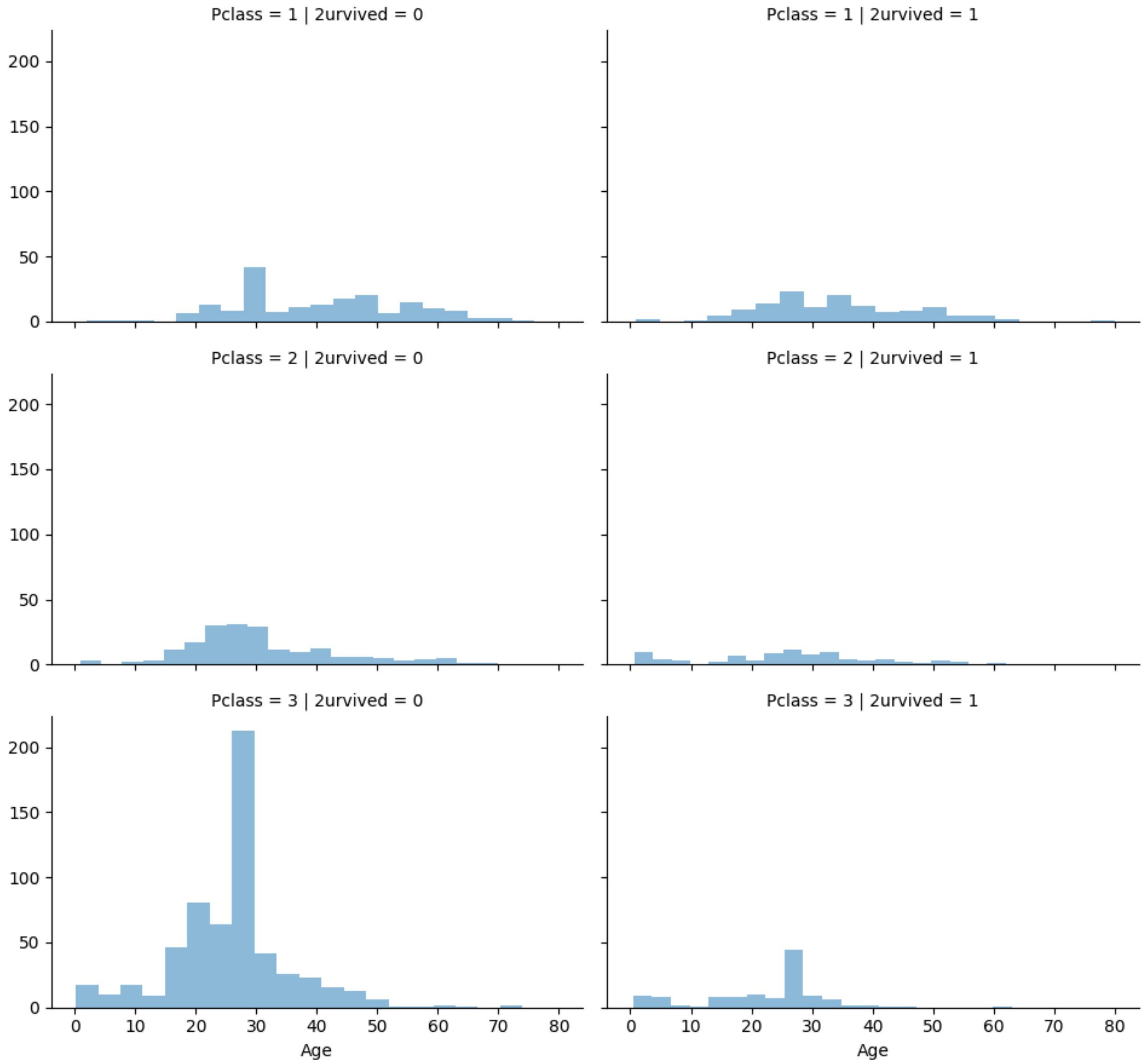
	sibsp	2urvived
1	1	0.351097
2	2	0.309524
0	0	0.235690
3	3	0.200000
4	4	0.136364
5	5	0.000000
6	8	0.000000

In [27]:
```
g = sns.FacetGrid(data, col='2urvived')
g.map(plt.hist, 'Age', bins=20)
```

Out[27]: <seaborn.axisgrid.FacetGrid at 0x7ecd59dbf320>



In [29]:
```
grid = sns.FacetGrid(data, col='2urvived', row='Pclass', aspect=1.6)
grid.map(plt.hist, 'Age', alpha=.5, bins=20)
grid.add_legend();
```

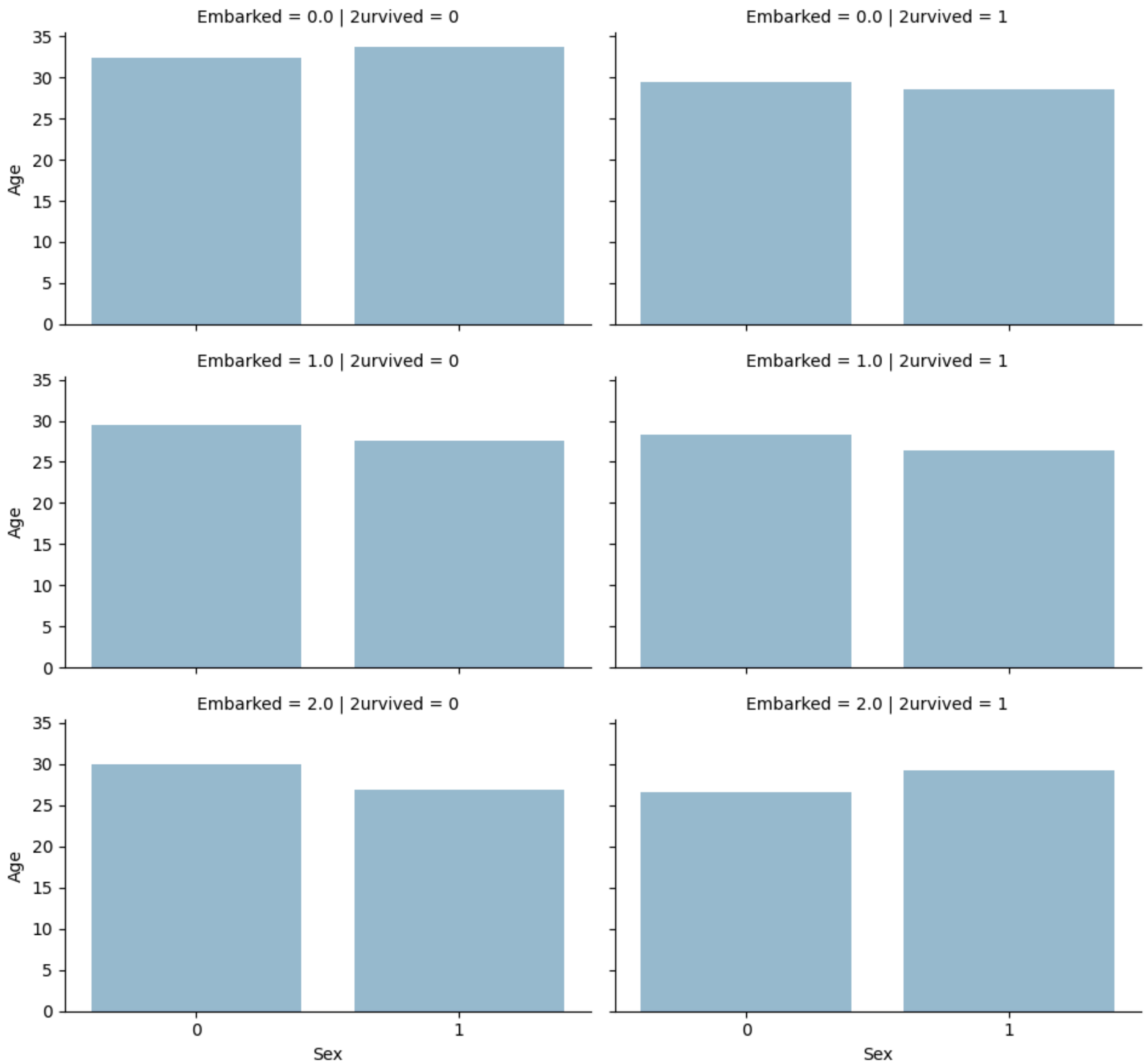


In [30]:

```
 grid = sns.FacetGrid(data, row='Embarked', col='2urvived', aspect=1.6)
grid.map(sns.barplot, 'Sex', 'Age', alpha=.5, ci=None)
grid.add_legend()
```

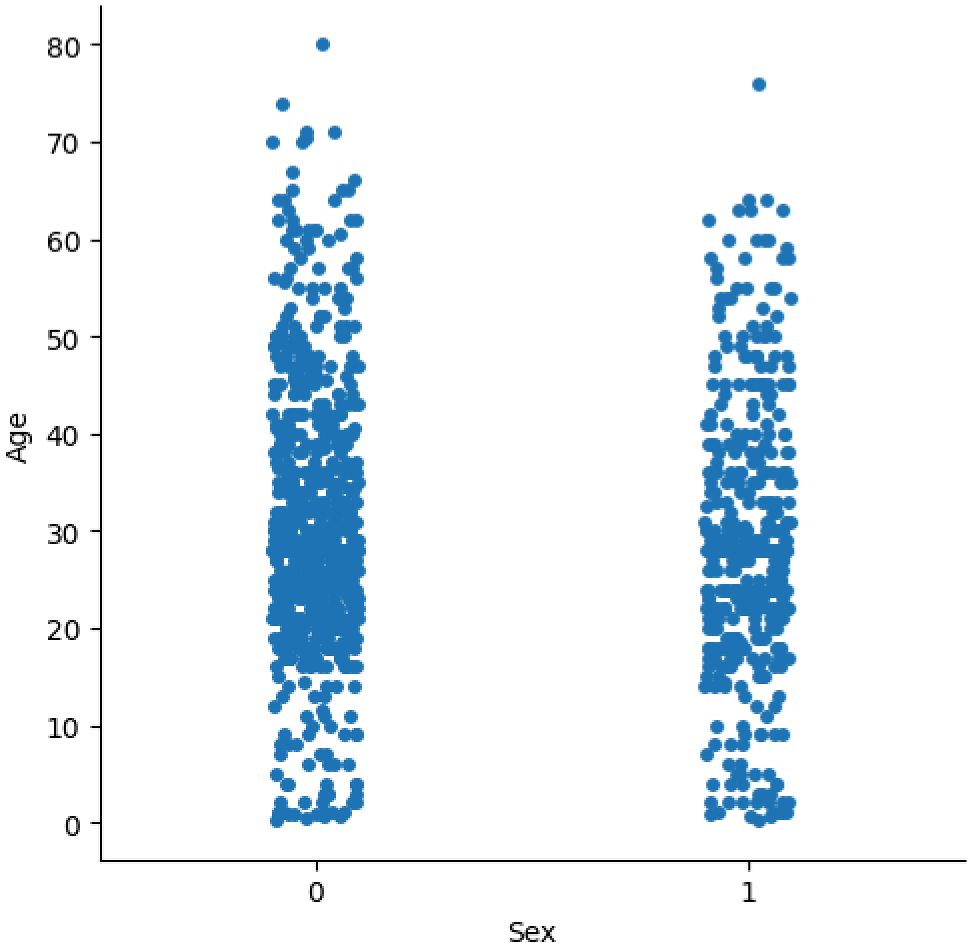
```
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:718: UserWarning: U
sing the barplot function without specifying `order` is likely to produce an in
correct plot.
  warnings.warn(warning)
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:854: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  func(*plot_args, **plot_kwargs)
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:854: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  func(*plot_args, **plot_kwargs)
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:854: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  func(*plot_args, **plot_kwargs)
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:854: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  func(*plot_args, **plot_kwargs)
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:854: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  func(*plot_args, **plot_kwargs)
/usr/local/lib/python3.12/dist-packages/seaborn/axisgrid.py:854: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  func(*plot_args, **plot_kwargs)
```

```
Out[30]: <seaborn.axisgrid.FacetGrid at 0x7ecd567be870>
```



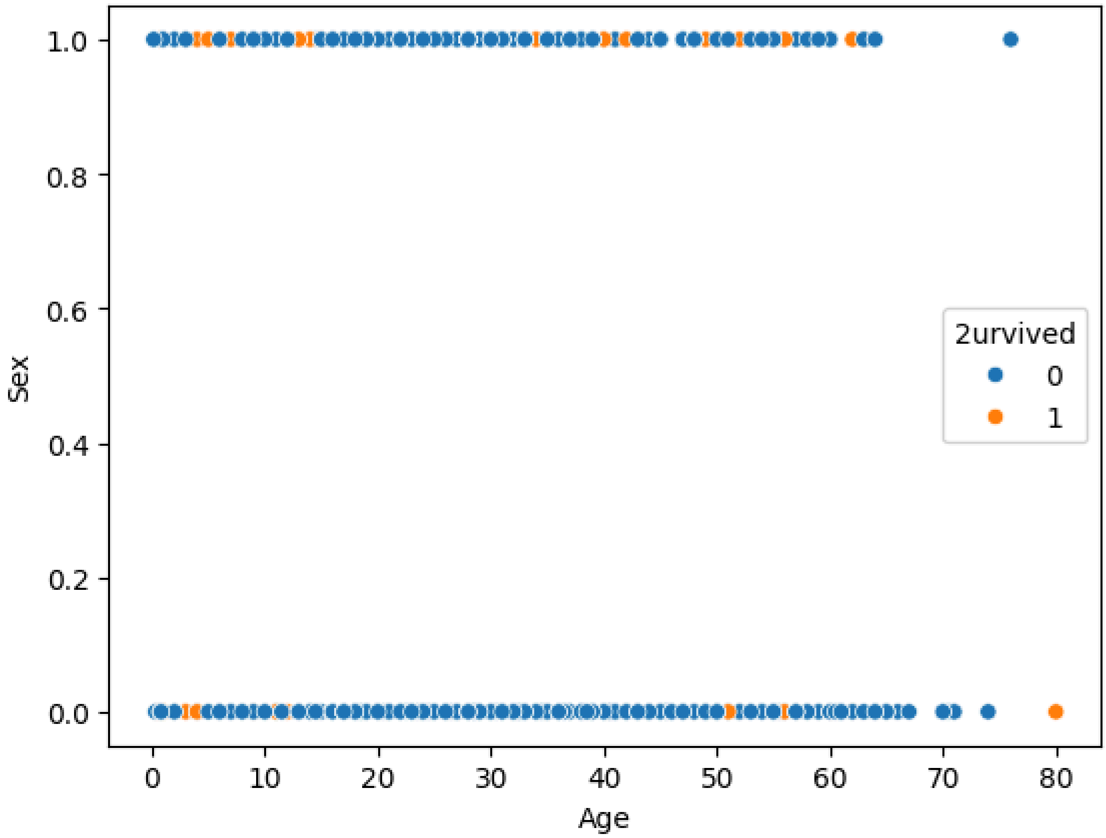
In [31]:
```
sns.catplot(x= 'Sex', y = 'Age', data=data, kind = 'strip')
```

Out[31]: <seaborn.axisgrid.FacetGrid at 0x7ecd565c5970>



In [32]:
```
sns.scatterplot(x = 'Age', y = 'Sex', hue = '2urvived', data = data)
```

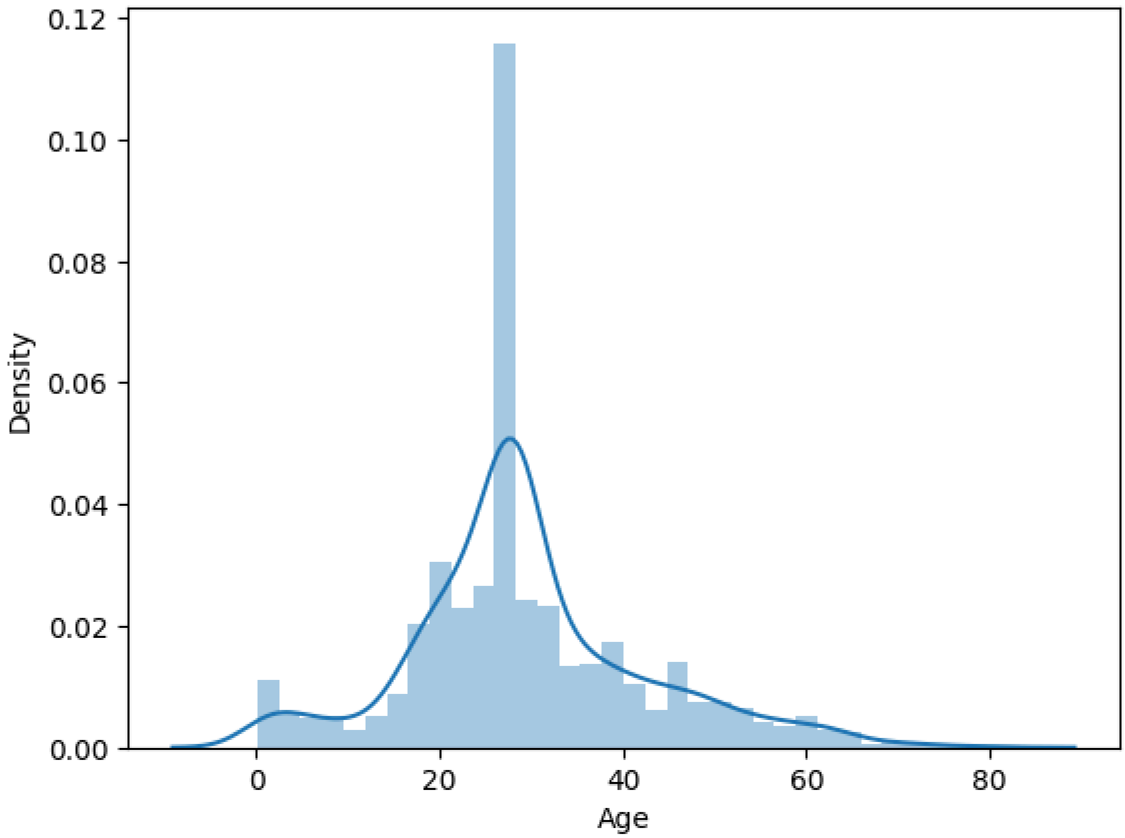
Out[32]:
```
<Axes: xlabel='Age', ylabel='Sex'>
```



In [33]:
```
sns.distplot(data['Age'])
```

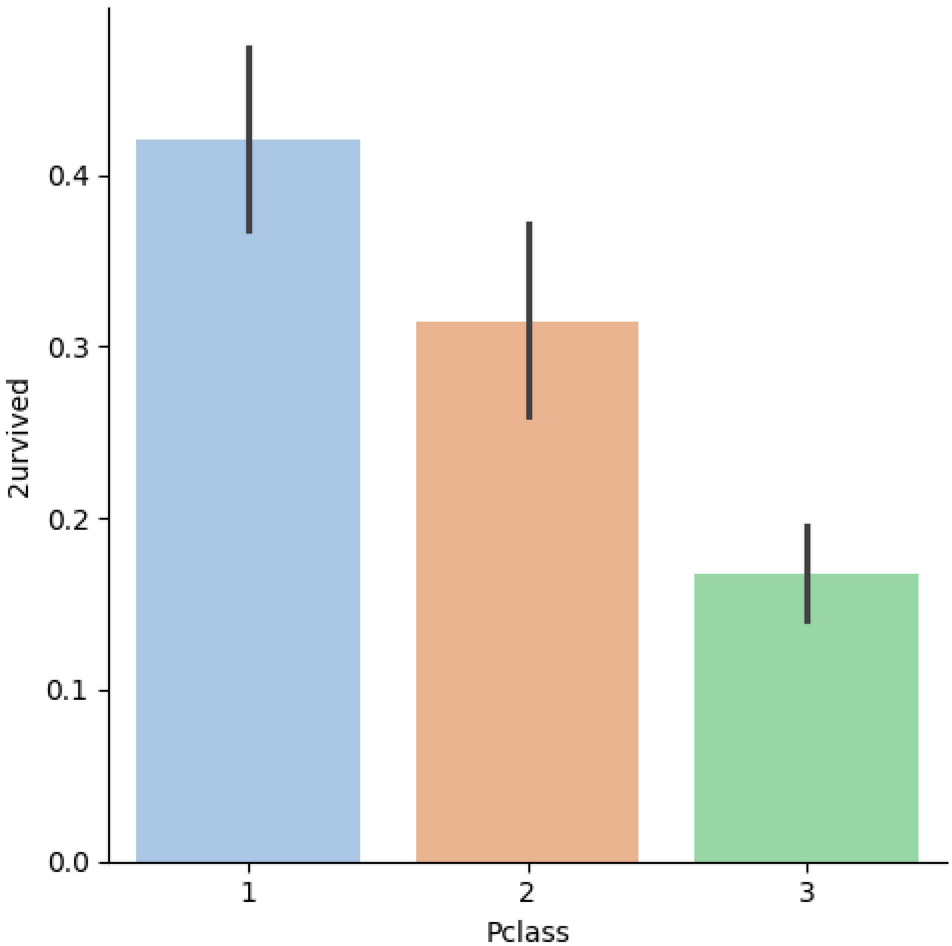
```
/tmp/ipykernel_6034/2317092479.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(data['Age'])
```

Out[33]:
```
<Axes: xlabel='Age', ylabel='Density'>
```



In [36]:
```
sns.catplot(x='Pclass', y='2urvived', data=data, kind='bar', hue='Pclass', pal
```

Out[36]:
```
<seaborn.axisgrid.FacetGrid at 0x7ecd59eaf440>
```



In [37]:
```
data[['Pclass', '2urvived']].groupby(['Pclass'], as_index=False).mean()
```

Out[37]:

	Pclass	2urvived
0	1	0.421053
1	2	0.314079
2	3	0.167842

In [38]:
```
data[['Age', '2urvived']].groupby(['Age'], as_index=False).mean()
```

Out[38]:

	Age	2urvived
0	0.17	0.000000
1	0.33	0.000000
2	0.42	1.000000
3	0.67	1.000000
4	0.75	0.666667
...	...	...
93	70.50	0.000000
94	71.00	0.000000
95	74.00	0.000000
96	76.00	0.000000
97	80.00	1.000000

98 rows × 2 columns

In [39]:

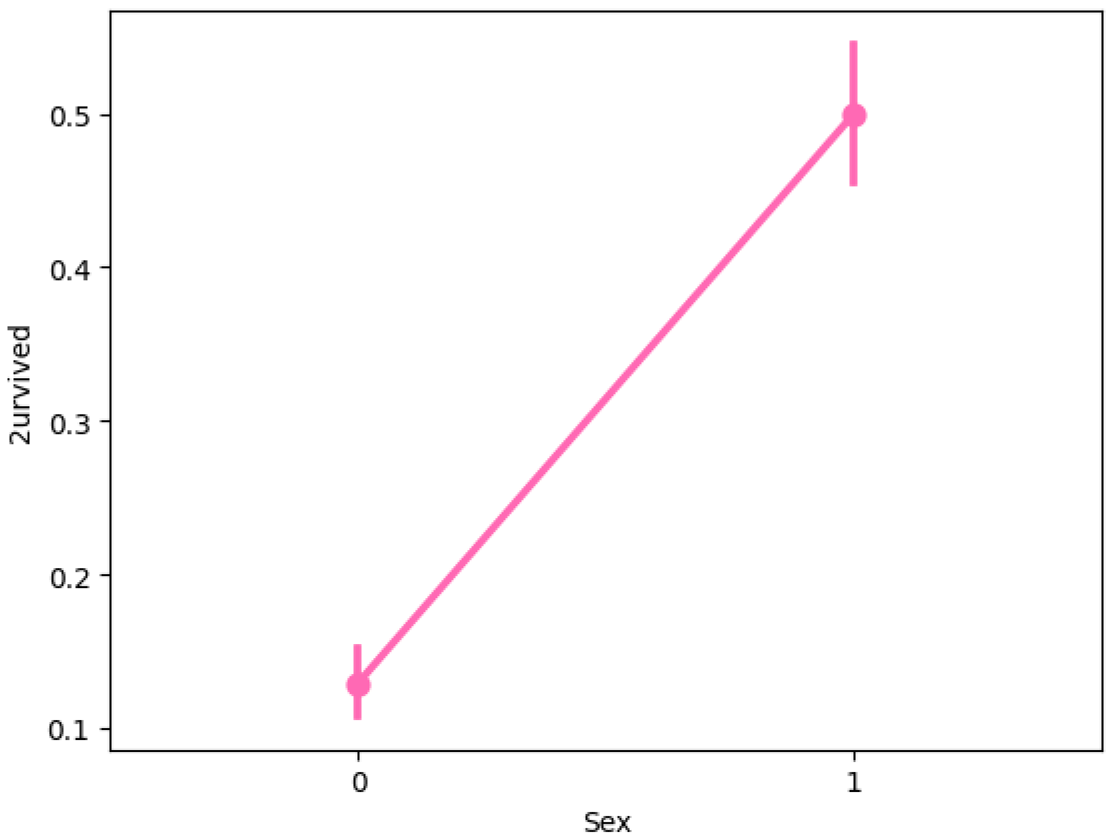
```
data[['Sex', '2urvived']].groupby(['Sex'], as_index=False).mean()
```

Out[39]:

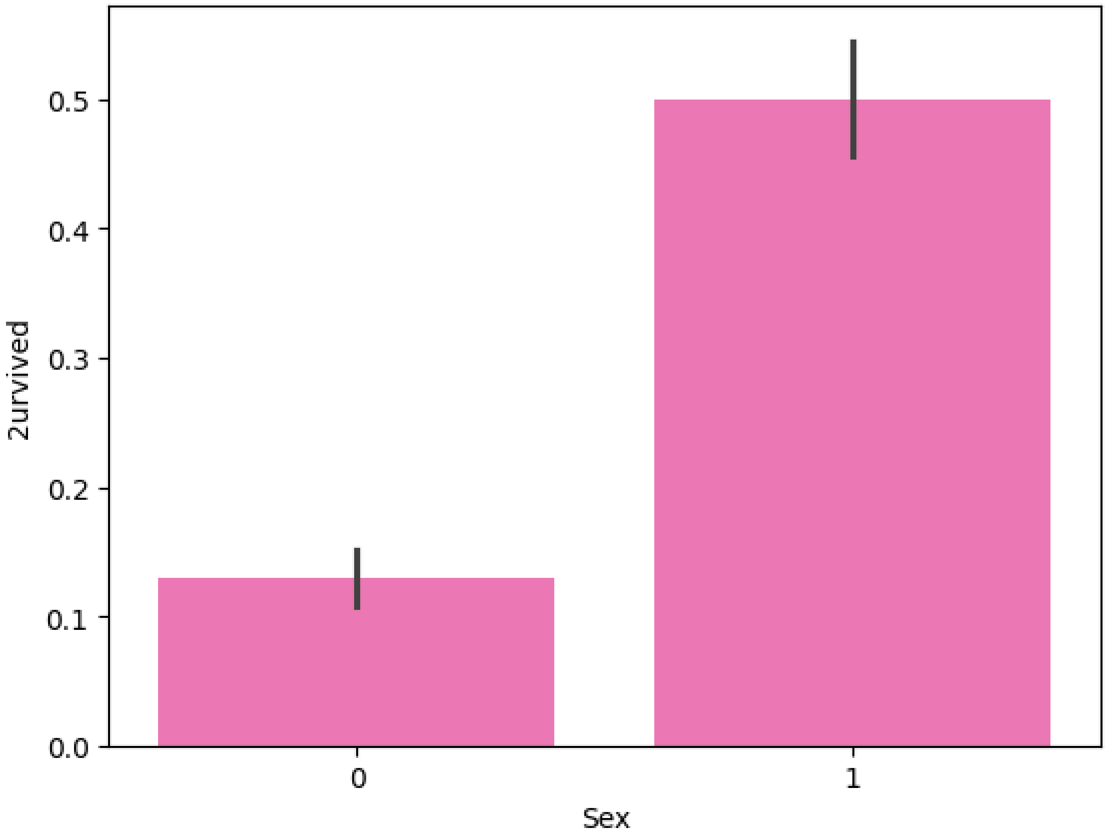
	Sex	2urvived
0	0	0.1293
1	1	0.5000

In [41]:

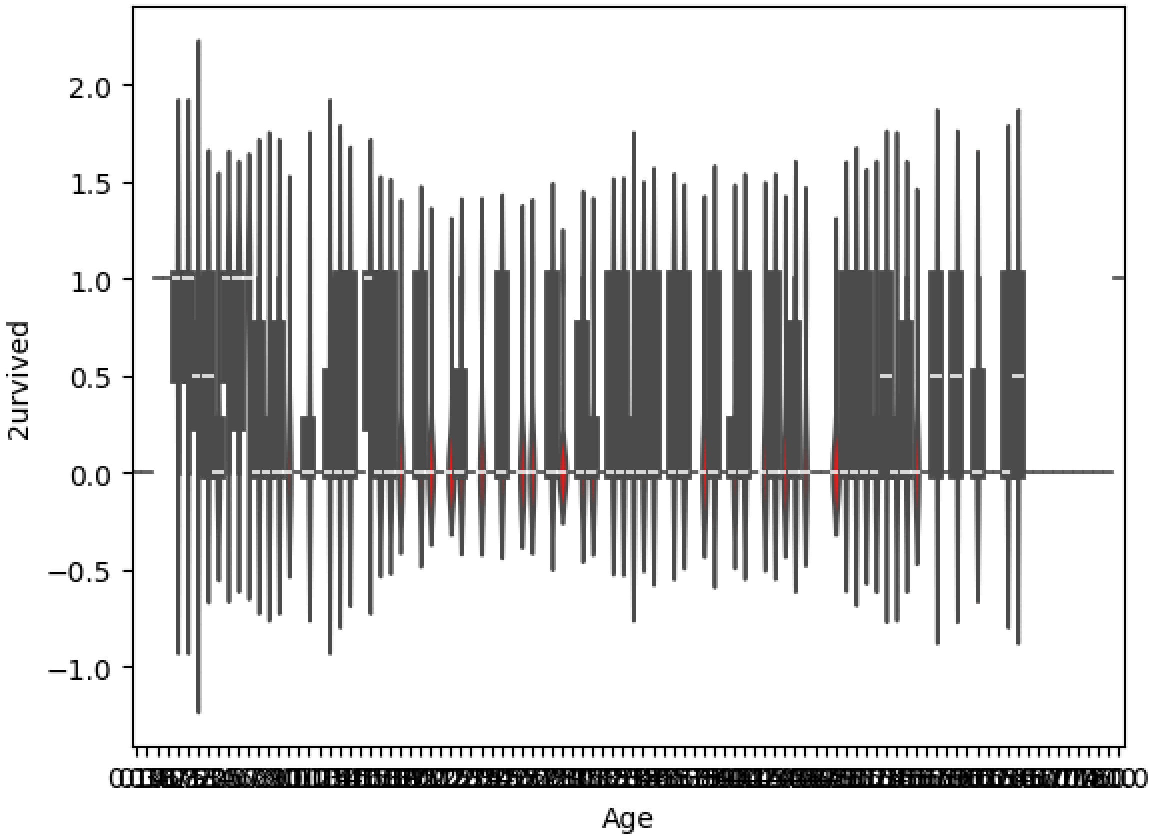
```
sns.pointplot(x = "Sex", y = "2urvived",  data = data, color='#FF69B4')
plt.show()
```



In [42]:
```
 sns.barplot(x = "Sex", y = "2urvived", data = data, color='#FF69B4' )
plt.show()
```

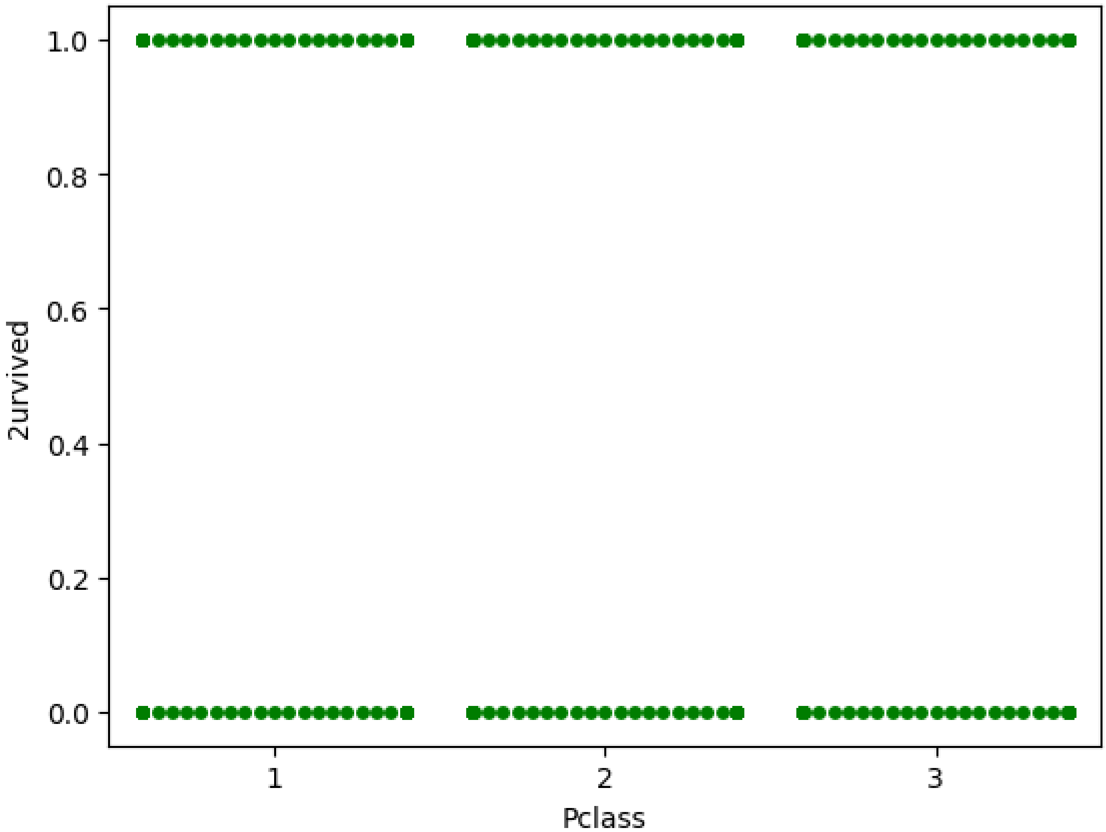


In [45]:
```
sns.violinplot(x = "Age", y = "2urvived",data=data, color='Red' )
plt.show()
```



In [46]:
```
sns.swarmplot(x = "Pclass", y ="2urvived", data = data, color='Green')
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarnin
g: 84.5% of the points cannot be placed; you may want to decrease the size of t
he markers or use stripplot.
  warnings.warn(msg, UserWarning)
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarnin
g: 81.9% of the points cannot be placed; you may want to decrease the size of t
he markers or use stripplot.
  warnings.warn(msg, UserWarning)
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarnin
g: 92.9% of the points cannot be placed; you may want to decrease the size of t
he markers or use stripplot.
  warnings.warn(msg, UserWarning)
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarnin
g: 88.2% of the points cannot be placed; you may want to decrease the size of t
he markers or use stripplot.
  warnings.warn(msg, UserWarning)
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarnin
g: 86.3% of the points cannot be placed; you may want to decrease the size of t
he markers or use stripplot.
  warnings.warn(msg, UserWarning)
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarnin
g: 94.6% of the points cannot be placed; you may want to decrease the size of t
he markers or use stripplot.
  warnings.warn(msg, UserWarning)
```



In [ ]: